

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The UDP protocol reveals that the source IP (192.51.100.15:52444) is requesting the IP address of yummyrecipesforme.com from the DNS server at 203.0.113.2. This is based on the results of the network analysis, which show that the ICMP response was a Destination Unreachable (Port Unreachable) message, returning the error 'udp port 53 unreachable.' The port noted in the error message is used for DNS, which is used for connecting users to digital resources like a website. The most likely issue is firewall restrictions preventing access, or there could be a DoS attack attempting to overload the server.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The time the incident occurred was 13:24:32.192571 (1:24pm). The IT team became aware of the incident due to customers being unable to access the website yummyrecipesforme.com, specifically reporting a "destination port unreachable" error. The IT team subsequently verified these complaints by attempting to visit the site and successfully reproducing the issue. The IT team utilized a packet sniffer tool called tcpdump to dissect the logs retrieved and identify the root cause. The IT team found that the server was online but the DNS port was unreachable ("udp port 53 unreachable") for users. The most likely cause(s) of the incident are the firewall restrictions preventing access, or there could be a DoS attack attempting to overload the server.

Raw Log Evidence:

```
13:24:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?  
yummyrecipesforme.com. (24)
```

```
13:24:36.098564 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2  
udp port 53 unreachable length 254
```

```
13:26:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?  
yummyrecipesforme.com. (24)
```

```
13:27:15.934126 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2  
udp port 53 unreachable length 320
```

```
13:28:32.192571 IP 192.51.100.15.52444 > 203.0.113.2.domain: 35084+ A?  
yummyrecipesforme.com. (24)
```

```
13:28:50.022967 IP 203.0.113.2 > 192.51.100.15: ICMP 203.0.113.2  
udp port 53 unreachable length 150
```

Resolution and status:

Currently the issue remains unresolved as of this report. The next steps that should be taken are to review the firewall configurations to identify if that is the root issue, then the IT team should monitor the incoming network traffic to see if a DoS attack is present and ongoing